

COURSE TITLE : POWER PLANT INSTRUMENTATION
COURSE CODE : 5215
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Introduction to power plants	14
2	Measurement and control in Boiler	14
3	Measurement and control in Air-Fuel	14
4	Measurement and Control in turbine	10
TOTAL		52

COURSE OUTCOME

Module	G.O	On completion of the study of this Module the student will be able
1	1	To understand the various electrical energy Sources
	2	To Comprehend the steam power plant
	3	To Comprehend the nuclear power plant
	4	To know the efficiency of thermal power plant
	5	To Comprehend the different elements of thermal power plant
	6	To understand the main flow circuits of thermal power plant
2	1	To understand the measurements in water circuit of Boiler
	2	To comprehend the control in water circuit of Boiler
	3	To know the impurities in feed water
	4	To know the feed water treatment
	5	To Understand Interlocks for boiler Operation.
3	1	To comprehend the measurements in Air- fuel circuit of Boiler
	2	To comprehend the control in Air -fuel circuit of Boiler
	3	To know the optimization of combustion control
	4	To understand furnace draft control
	5	To comprehend the various analytical measurement in power plant
4	1	To Understand the classification of turbine, based on process conditions
	2	To comprehend the measurement system in Turbine
	3	To comprehend the control system in Turbine
	4	To understand the lubrication system for turbo -alternator
	5	To understand the Master control for power station

On completion of the study the student will be able

MODULE I INTRODUCTION TO POWER PLANT

1.1.0 To understand the Various electrical energy Sources

- 1.1.1 To list the Various Non –conventional energy Sources.
- 1.1.2 To list the Various conventional energy Sources..
- 1.1.3 To differentiate the non-conventional and conventional energy sources

1.2.0 To Comprehend the steam power plant

- 1.2.1 To explain the layout of typical thermal power plant.
- 1.2.2 To describe the generation of electricity in thermal power plant.

1.3.0 To Comprehend the nuclear power plant

- 1.3.1 To explain the block diagram of a nuclear power plant
- 1.3.2 To list the main components of nuclear reactor
- 1.3.3 To define Fusion
- 1.3.4 To define Fission
- 1.3.5 To list the advantages and disadvantages of nuclear power

1.4.0 To know the efficiency of thermal power plant

- 1.4.1 To list main components of thermal power plant
- 1.4.2 To define the boiler efficiency, turbine efficiency and alternater efficiency.

1.5.0 To Comprehend the different elements of thermal power plant

- 1.5.1 To explain the various equipments of a thermal power plant

1.6.0 To understand the main flow circuits

- 1.6.1 To list the flow circuit of thermal power plant
- 1.6.2 To explain the Feed water and steam circuit
- 1.6.3 To explain the cooling water circuit
- 1.6.4 To describe the Fuel-ash circuit
- 1.6.5 To explain the Air flue gas circuit

MODULE II MEASUREMENT AND CONTROL IN BOILER

2.1.0 To understand the measurements in water circuit of Boiler

- 2.1.1 To explain the boiler feed water circulation
- 2.1.2 To describe the measurements in water circuit.

2.2.0 To comprehend the control in water circuit of Boiler

- 2.2.1 To explain the Boiler Drum level control
- 2.2.2 To describe the superheated steam temperature control
- 2.2.3 To explain the steam pressure control

2.3.0 To know the impurities in feed water

- 2.3.1 To explain the impurities of feed water
- 2.3.2 To describe the effects of impurities of water
- 2.3.3 To explain the measurements of impurities of water

2.4.0 To know the feed water treatment

2.4.1 To describe the external treatment of feed water treatment

2.4.2 To explain the internal treatment of feed water treatment

2.5.0 To understand the Interlocks for Boiler operation

2.5.1 To explain the safety inter locks

2.5.2 To describe the start –up and shutdown interlocks

2.5.3 To explain sequencing of Burner and other operations

MODULE III MEASUREMENTS AND CONTROL IN AIR –FUEL

3.1.0 To comprehend the measurements in Air- fuel circuit of Boiler

3.1.1 To list the types of Fuels used in boiler

3.1.2 To know the combustion

3.1.3 To list the constituents of flue gases and waste gases

3.1.4 To list the measurements in air fuel circuit

3.2.0 To comprehend the control in Air –fuel circuit of Boiler

3.2.1 To explain the combustion control with gas fuel (metered cross limited air –fuel ratio control)

3.2.2 To describe the combustion control with liquid fuel.

3.2.3 To explain the coal –air fuel ratio control

3.3.0 To know the optimization of combustion control

3.3.1 To describe the optimization of combustion control

3.4.0 To understand furnace draft control

3.4.1 To explain forced draft fan control, induced draft fan control and Balanced draft control.

3.5.0 To comprehend the various analytical measurement in power plant

3.5.1 To describe the Zirconia oxygen analyser

3.5.2 To explain the Combustibles(Co+H₂) analyser

3.5.3 To describe the Smoke detector

3.5.4 To explain the Dust monitor

MODULE IV MEASUREMENTS AND CONTROL IN TURBINE

4.1.0 To Understand the classification of turbine- based on process conditions

4.1.1 To describe the working of a Turbine

4.1.2 To list the classification of turbine-based on process condition

4.2.0 To comprehend the measurements in Turbine

4.2.1 To explain the Turbine steam inlet system

4.2.2 To list the various measurements in Turbine

4.2.3 To list the various process parameters

4.2.4 To list the mechanical parameters

4.2.5 To list the electrical parameters

4.3.0 To comprehend the controls in Turbine

4.3.1 To list the turbine control system

- 4.3.2 To describe the safety control systems
- 4.3.3 To list the measurements in safety control system
- 4.3.4 To describe the process control systems

4.4.0 To understand the lubrication system for turbo –alternator

- 4.4.1 To explain the lubrication system for turbo –alternator

4.5.0 To understand the Master control for power station

- 4.5.1 To explain the master control for a power station

CONTENT DETAILS

MODULE I

List and compare various Non conventional energy Sources - steam power plant - layout of typical thermal power plant - generation of electricity in thermal power plant - block diagram of a nuclear power plant - main components of nuclear reactor - define fusion and fission- advantages and disadvantages of nuclear power plant - efficiency of thermal power plant - main components of thermal power plant - define the boiler, turbine alternator efficiency - elements and equipments of thermal power plant - flow circuits of thermal power plant.

MODULE II

Boiler feed water measurements and control – pressure – temperature - flow – boiler drum level measurement and control –drum level indicator local and remote - two element control – three element control – five element control – impurities and its effects in feed water – feed water treatment - external and internal treatment - safety interlocks for Boiler operation.

MODULE III

Types of Fuels used in boiler - constituents of flue gases & waste gases - measurements in air fuel circuit - the combustion control with gas fuel - metered cross limited air –fuel ratio control - combustion control with liquid fuel - coal –air fuel ratio control - optimization of combustion control - furnace draft control forced draft control – induced draft control - Balanced draft control - Zirconia oxygen analyzer - the Combustibles(Co+H₂) analyzer - Smoke detector - Dust monitor.

MODULE IV

Turbine- working - classification of turbine - measurements in Turbine – process – mechanical and electrical parameters – Turbine controls - lubrication system for turbo –alternator- Master control for power station.

REFERENCES

1. K.Krishnaswamy and M.Ponnibala- Power Plant Instrumentation- PHI Learning Pvt Ltd, 2011
2. B.G.Liptak-Instrumentation Engineers Handbook (Process Measurement and Analysis), Fourth Edition- Chilton Book Co, 2003
3. John G Webster-The Measurement, Instrumentation, and Sensors Handbook- CRC and IEEE Press, 1999.
4. Sam. G . Dukelow- Control of Boilers- Instrument Society of America